CGA: Curvature-Guided Attention for Remote Sensing Image Super-Resolution

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***Abstract*—Remote sensing (RS) image super-resolution (SR) supports downstream tasks in mapping, monitoring, and recognition; yet, curvilinear structures and repetitive textures that encode scene topology are aliased by downsampling. Generic attention without geometric priors tends to yield jagged edges and washed-out detail. To address this, we present curvature-guided attention (CGA), which injects curvature cues into local windowed attention and token-based global aggregation within a standard transformer framework. A local curvature-guided attention (LCGA) reinforces edge and line continuity inside windows, while a curvature-guided token attention (CGTA) selects globally informative tokens with near linear complexity, avoiding the heavyweight global modules and extra branches. The resulting design preserves structural layout and texture regularity without degrading natural regions and integrates with conventional residual groups (RGs). Across datasets and scale factors, CGA attains higher reconstruction fidelity than state-of-the-art SR models, with qualitative results showing clearer geometry and fewer artifacts. Ablations isolate the role of each component and confirm complementary benefits, and an efficiency-oriented variant retains the main trend under tighter memory and latency. We further validate CGA on real-world RS imagery without simulated degradation. These findings indicate that curvature guidance is an effective and practical inductive bias for RS image SR, strengthening both aggregate and class-level performance across scenes and scales while keeping computational complexity tractable. Code and pretrained models are available at: https://github.com/mwaleedaslam/CGA**

***Index Terms*—Curvature-guided attention (CGA), deep learning, remote sensing (RS) image, super-resolution (SR), transformer.**

I. Introduction

SINGLE image super-resolution (SR) has advanced rapidly from early convolutional models to attention-based transformers, yielding strong pixel-level fidelity and perceptual quality on natural images [1], [2], [3], [4], [5]. In remote sensing (RS), however, downstream geographic information systems (GIS) tasks depend critically on curvilinear structures e.g., road centerlines, rivers, coastlines, roof edges, and utility corridors whose topology and local geometry must be preserved for mapping, change detection and small-object analysis [6]. Standard objectives and metrics can underemphasize these ridge-like primitives, so methods that excel on textures may still distort curvature and alignment, degrading GIS utility despite competitive averages [7]. This leaves a gap between the perceptual quality and the structural accuracy required by RS applications.

To ground this need at the output level, Fig. 1(a) visualizes one building scene where we overlay ridge evidence directly on the image: high-resolution (HR) ridge outline (blue), SR near-aligned segments (green), and SR geometric displacements beyond a small tolerance (red). We summarize this with a single scalar named curvature integrity score (CIS) that rewards HR ridge pixels both for being spatially close to SR ridges and for forming unbroken runs along the curve. It means, for each HR ridge pixel x, let $d(x)$ be its distance to the nearest SR ridge and let $\mathbb{1}_{\text{run}}(x)$ be that x lies in a covered contiguous segment of length at least L_0; then

$$\text{CIS} = \mathop{\mathbb{E}}_{x\in\text{HR ridge}}\left[\exp\left(-\frac{d(x,E_{\text{SR}})^2}{2\sigma^2}\right)\mathbb{1}_{\text{run}}(x)\right]\in[0,1]$$

where E_{SR} is the SR edge set, σ and the run-length threshold L_0 control spatial tolerance and continuity. In practice, HR ridge bands are extracted with multiscale edge detection and simple morphological cleanup, and distances are computed with a standard Euclidean distance transform [8], [9], [10], [11]. In our example, CIS agrees with the visual overlay by preferring reconstructions that reduce breakages and displacements.

Furthermore, Fig. 1(b) examines *how attention is allocated on ridge neighborhoods*. We restrict analysis to the HR ridge band and compare three variants of attention: standard (Std), curvature-guided (Curv), and their learned blend (Blend). Two concise diagnostics summarize behavior: 1) the ridge-band attention ratio (RBAR)

$$\text{RBAR}_X = \frac{\sum_{\text{ridge band}} A_X}{\sum_{\text{image}} A_X},\quad \text{for } X\in\{\text{Std},\text{Curv},\text{Blend}\}$$

which measures the fraction of a variant's attention mass that concentrates on ridge pixels and 2) the ridge-band attention

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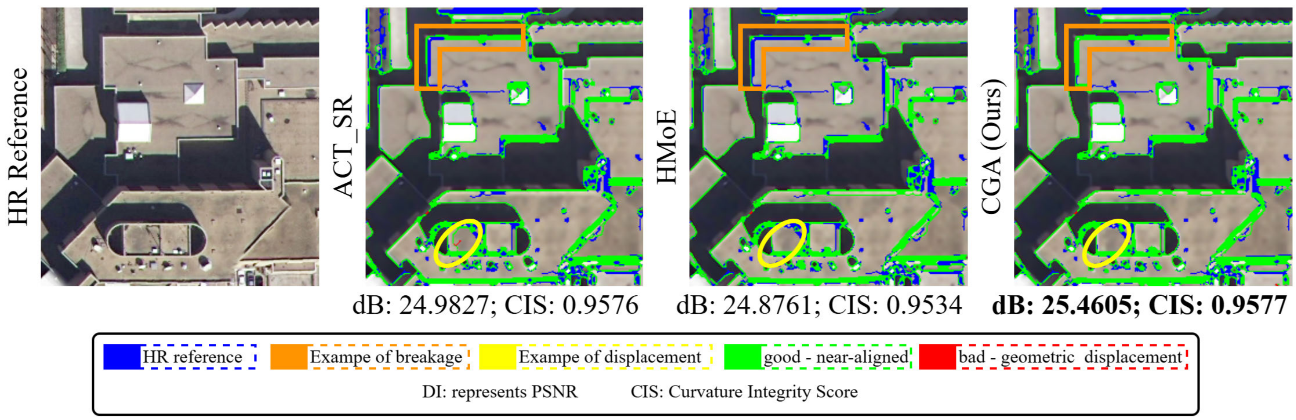
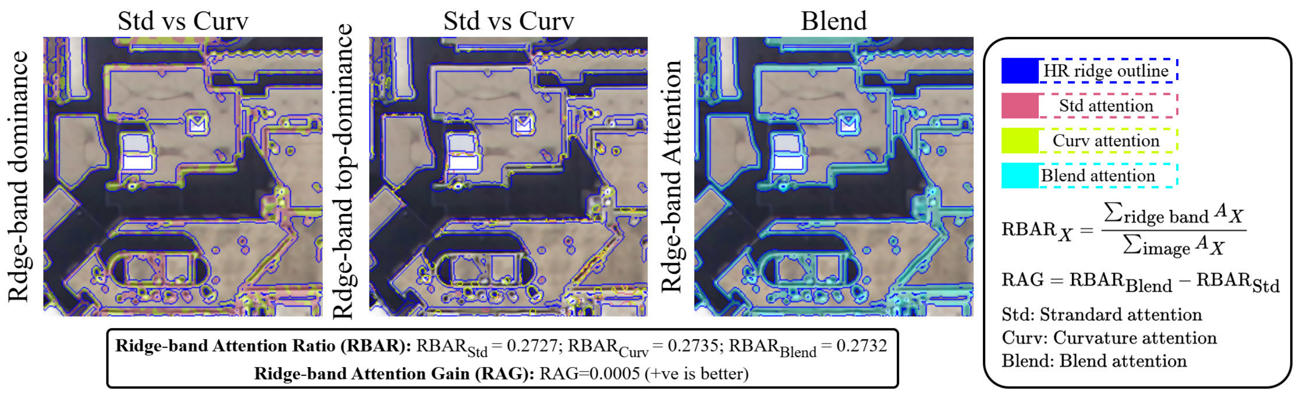


(a)



(b)

Fig. 1. Curvilinear structure diagnostics. (a) Curvilinear alignment and breakages (CABs) on ridge outlines. The HR ridge contour is shown in blue. Near-aligned SR segments are shown in green. Geometric displacement vectors are shown in red. CIS↑ summarizes ridge proximity and continuity. Insets: corner case in orange and a displacement case in yellow. (b) Ridge-band attention diagnostics evaluated inside the HR ridge band. The dominance map compares Std and Curv attention. The hotspot map shows the union of the top 15% dominant responses. The blended overlay supports visual inspection. RBAR reports the fraction of attention inside the ridge band. RAG reports ridge-band attention gain, where higher values indicate a better performance.

gain (RAG)

$$\mathrm{RAG} = \mathrm{RBAR}_{\text{Blend}} - \mathrm{RBAR}_{\text{Std}}.$$

RAG, therefore, measures the net increase in ridge-focused attention achieved by the blended variant relative to standard attention. Qualitatively, the ridge-band dominance map (Std versus Curv) highlights that Curv dominates precisely at curvature-rich loci such as roof corners and tight bends, while Std may prevail on relatively straight, weakly curved segments. The top-dominance/hotspot map (top 15% union within the band) isolates the strongest attention maxima, which cluster on those same curvature loci, indicating that curvature cues concentrate attention where geometric continuity is fragile. The separate Blend overlay shows that Blend inherits this ridge focus from Curv while remaining stable off-band. Quantitatively, this behavior is reflected by RBAR (Curv $\geq$ Std and Blend close to Curv on the band) and a positive RAG, consistent with prior practice that uses attention mass as a region-level diagnostic in transformer models [12], [13]. These observations motivate curvature fidelity as a design target.

Convolutional neural networks (CNNs) [1], [2], [14] are efficient, but their locality makes long-range association along extended curves difficult to capture without very deep stacks. Recent transformer variants [3], [15], [16] strengthen nonlocal modeling; yet, they do not explicitly reweight attention toward ridges, and quadratic all-to-all attention is prohibitive for large RS images. Efficiency-oriented designs such as windowed or token-selective attention improve scalability [17], [18], [19], but purely windowed mechanisms remain bounded by context size; enlarging or overlapping windows increases cost and still limits global reach [4], [20]. Consequently, there is a methodological gap: scalable formulations that explicitly maintain curvature fidelity on ridges, where RS tasks are most sensitive.

To address this, we propose curvature-guided attention (CGA): a geometry-aware SR architecture that encodes curvature in attention while remaining scalable. CGA couples two complementary mechanisms. A local curvature-guided attention (LCGA) module operates within bounded windows and applies a learned blend of curvature-modulated and standard attention, concentrating mass on ridge neighborhoods without destabilizing nonridge regions. A curvature-guided token attention (CGTA) mechanism enables efficient global interaction by routing queries to a compact, content-adaptive token set selected via curvature-aware saliency, yielding linear-like behavior with respect to image size while maintaining

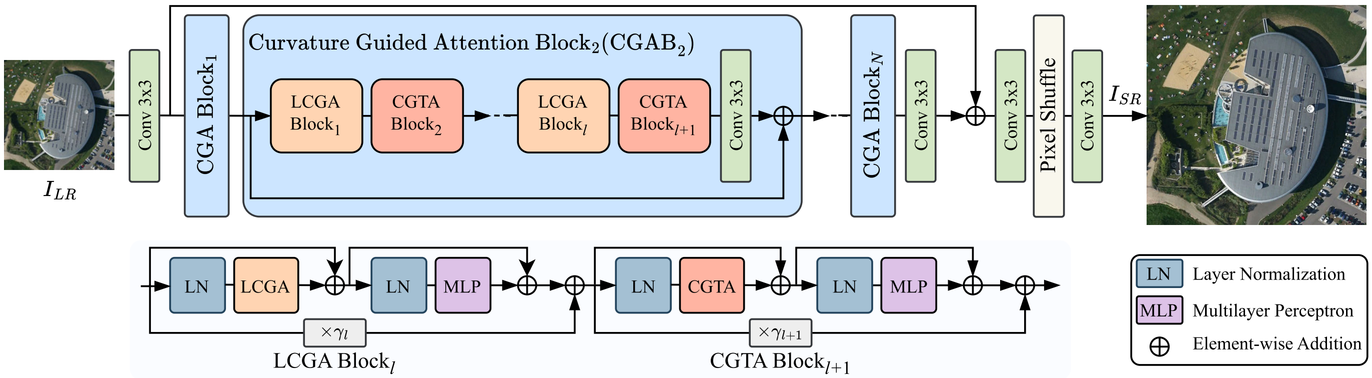


Fig. 2. Overview of the CGA architecture. LCGA blocks and CGTA blocks are alternated. γ_l is a learnable adaptor of the lth block for adaptive integration and stabilized training.

long-range geometric coherence. Together, LCGA preserves local edge topology and mitigates attention leakage into textured but nonridge areas, whereas CGTA propagates structure along extended curves. To the best of our knowledge, curvature cues have not previously been used to guide both local window attention and token selection for RS single-image SR. Empirically, across public RS image SR benchmarks, CGA delivers strong quantitative gains and sharper, more coherent ridge geometry in visual comparisons.

This article makes the following contributions.

1) We introduce CGA, a geometry-aware SR architecture that explicitly emphasizes curvature to preserve ridge geometry while retaining stability off-ridge, achieving competitive or state-of-the-art results on multiple public RS image SR datasets in both accuracy and visual quality.
2) A local module LCGA that blends curvature modulated and standard attention to emphasize ridge neighborhoods within windows while preserving the stabilizing effect of standard attention in nonridge regions.
3) A token-based global mechanism CGTA that selects a compact, content-adaptive set via curvature aware saliency to realize linear-like interactions across the image and to improve the structural alignment along long-range curves.

The remainder of this article is organized as follows. Section II reviews related works. Section III presents the proposed CGA, including LCGA and CGTA. Section IV reports the experimental setup, including datasets, implementation, evaluation metrics, ablation study, and result details. Section V provides discussion and Section VI concludes this article.

II. Related Works

A. Curvature and Ridges in Vision

Scale-space theory provides a principled basis for multiscale geometry [21]. Level-set curvature computed on smoothed intensities offers a contrast-robust differential invariant for geometric processing [22]. Ridges and valleys can be localized with subpixel accuracy using Hessian-based detectors [23] and enhanced via vesselness-style measures that aggregate eigeninformation across scales [24]. In our work these tools serve strictly as analysis constructs: HR ridge bands and centerlines are used to visualize SR geometric deviations and to quantify attention allocation through RBAR and RAG in Fig. 1(b). They frame the ridge-centric failure modes that CGA is designed to address.

B. Efficient Attention for Large Images

Windowed and shifted self-attention preserve locality and hardware efficiency [4], [20], but enlarging or overlapping windows still limits global association. Content-adaptive token mechanisms reduce the effective key and value set and approach linear scaling [25], [26], [27], [28], while low-rank and sparse approximations provide complementary scalability [29], [30], [31]. In RS image SR, token-selective designs show that sparsity can preserve fidelity at scene scale [17], [18], [19]; yet, they remain curvature-agnostic. Our perspective, validated qualitatively and quantitatively in Fig. 1, is to make curvature an explicit signal in attention. CGA blends curvature-guided and standard attention for local stability (LCGA) and routes queries to content-adaptive tokens for scalable global association (CGTA), directly targeting the ridge alignment and continuity highlighted by CIS and RBAR/RAG.

III. Methodology

In this section, we provide an overview of the CGA architecture and its two core components: LCGA and CGTA. We, then, present reconstruction detail; full designs of LCGA and CGTA are presented in Sections III-B and III-C, respectively.

A. Overview of CGA

As shown in Fig. 2, given an LR image $I_{\mathrm{LR}} \in \mathbb{R}^{H\times W\times 3}$, CGA first extracts shallow features with a 3×3 convolution

$$\mathcal{F}_0 = \mathcal{H}_{3\times 3}(I_{\mathrm{LR}}) \in \mathbb{R}^{H\times W\times C}$$

where H and W denote the image height and width in pixels, respectively, 3 denotes the RGB channels, C is the channel width, and $\mathcal{F}_0$ represents the shallow feature map provided to the attention blocks. Following [32], a single 3×3 stem is an efficient and well-behaved choice that preserves fine spatial cues for downstream attention blocks.

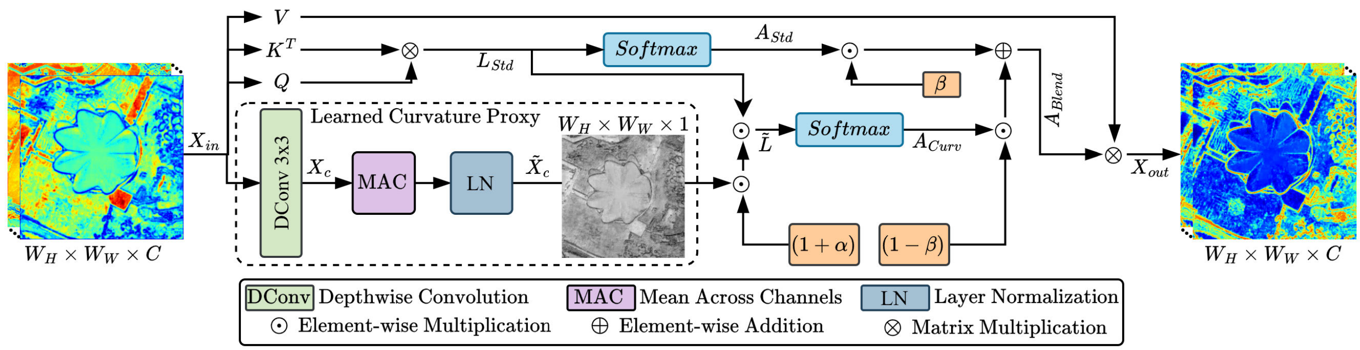


Fig. 3. LCGA. A depthwise 3×3 convolution with MAC $\rightarrow$ LN yields a curvature proxy $\tilde{X}_c$ that multiplicatively modulates standard window attention logits to produce $\tilde{L}$. The resulting A_{Curv} is blended with A_{Std} via a learned gate to form A_{Blend}, emphasizing ridge neighborhoods while preserving stability off ridge.

Deep features are then produced by a stack of N curvature-guided attention blocks (CGABs). Each CGAB alternates a local LCGA sublayer and a global CGTA sublayer in a prenorm design with an MLP [33] head; a lightweight, learnable adaptor γ stabilizes residual blending across blocks. Denoting the ith CGAB by $\mathcal{H}_i^{\text{CGAB}}(\cdot)$

$$\mathcal{F}_i = \mathcal{H}_i^{\text{CGAB}}(\mathcal{F}_{i-1}), \quad i = 1, \ldots, N$$
$$\mathcal{F}_{\text{DF}} = \mathcal{H}_{3\times3}(\mathcal{F}_N).$$

Here, $\mathcal{F}_i$ denotes the intermediate deep features after the ith CGAB, $\mathcal{F}_N$ denotes the deep features after the last CGAB, and $\mathcal{F}_{\text{DF}}$ denotes the poststack features obtained by a 3×3 consolidation convolution.

Finally, shallow and deep features are fused and reconstructed to RGB

$$I_{\text{SR}} = \mathcal{H}_{RC}\left(\mathcal{F}_0 + \mathcal{F}_{\text{DF}}\right) \in \mathbb{R}^{H' \times W' \times 3}$$

where $\mathcal{H}_{RC}(\cdot)$ is a lightweight reconstruction head and $H' \times W'$ is the SR size.

Instantiation and Rationale: CGA is instantiated on standard transformer restoration plumbing [34] and injects curvature guidance in both the local and global paths. Locally, LCGA performs windowed attention to stabilize aggregation [35], while explicitly strengthening ridge/curve continuity inside each window via curvature-modulated logits. Globally, CGTA follows a content-adaptive routing strategy [27], [28] that selects a compact Top-k token set guided by curvature-aware saliency and applies blended cross-attention only over the selected keys/values. This design targets two key needs of SR: preserving fine curvilinear structures locally and maintaining long-range consistency along extended ridges, while reducing the attention cost from $O(N^2)$ to $O(Nk)$, i.e., near linear for fixed k/N.

We train CGA end-to-end with the L_1 loss

$$\mathcal{L}_1 = \left\| I_{\text{SR}} - I_{\text{HR}} \right\|_1$$

a standard choice in SR that encourages sharp, artifact-free detail and is widely used in strong baselines [2], [3]; L_1 is also empirically preferable to L_2 for restoration due to its robustness to outliers and reduced over-smoothing [36].

B. Local Curvature-Guided Attention

As shown in Fig. 3, LCGA operates within rectangular windows extracted from the feature map. Given $X_{\text{in}} \in \mathbb{R}^{W_H \times W_W \times C}$, LCGA maps $X_{\text{in}} \rightarrow X_{\text{out}}$ by blending standard window attention with a curvature-modulated variant, so that aggregation is emphasized on curvilinear/ridge neighborhoods while remaining stable off-ridge.

1) Learned Curvature Proxy: We apply a depthwise 3×3 operator to the window input

$$X_c = \mathcal{H}_{3\times3}^{D}(X_{\text{in}})$$

then, take mean across channels and LayerNorm to obtain a normalized scalar proxy at each window location

$$\tilde{X}_c = \mathcal{H}_{\text{LN}}\left(\mathcal{H}_{\text{MAC}}(X_c)\right).$$

Intuitively, the depthwise 3×3 filters act as a learnable bank of local differential-like operators: with a small receptive field they can approximate oriented derivative and second-order patterns (e.g., bend/turn responses) in each channel, activating strongly where the feature field changes direction. The MAC aggregation encourages cross-channel consensus and suppresses channel-specific semantics, yielding a single structural saliency map. LayerNorm further stabilizes the scale across scenes, so high responses indicate relative curvature saliency rather than raw activation magnitude. As illustrated in Fig. 3, $\tilde{X}_c$ highlights ridge/curvilinear patterns that LCGA aims to preserve, while remaining lightweight and learned end-to-end. Unlike hand-crafted curvature operators such as LoG/Laplacian that are fixed and scale-sensitive, the curvature guidance in CGA is learned end-to-end from SR supervision, allowing the proxy to adapt to RS imagery statistics.

2) Standard Attention in the Window: We flatten the window into $M = W_H W_W$ tokens and compute standard attention using linear projections. The standard attention logits and weights are

$$L_{\text{Std}} = \frac{Q \otimes K^T}{\sqrt{C}}, \quad A_{\text{Std}} = \mathcal{H}_{\text{Softmax}}(L_{\text{Std}})$$

where $\otimes$ denotes the matrix multiplication and $\mathcal{H}_{\text{Softmax}}(\cdot)$ applies softmax along the key dimension. The resulting $A_{\text{Std}} \in \mathbb{R}^{M \times M}$ provides a stable baseline distribution for local aggregation.

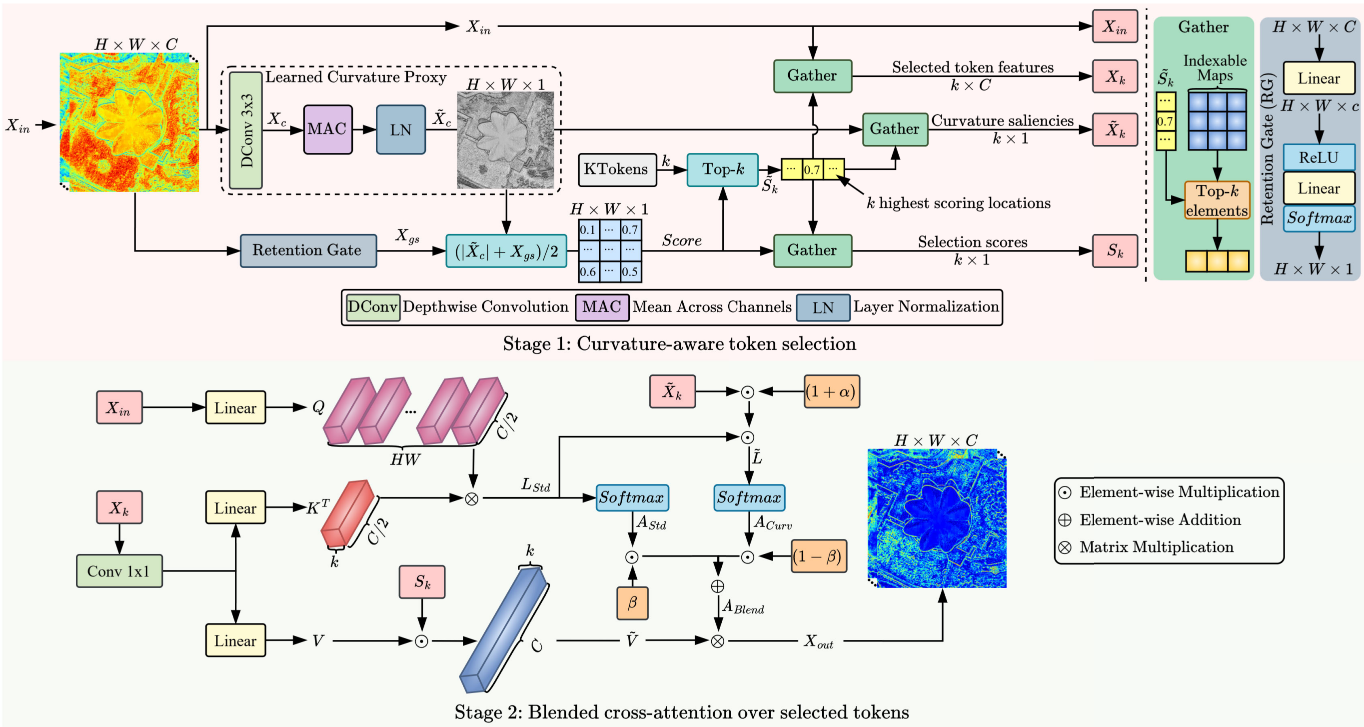


Fig. 4. CGTA with two stages. Stage 1 forms a selection score by combining curvature saliency $\tilde{X}_c$ and retention reliability X_{gs}, then applies a size-aware KTokens policy shown in (1) to choose Top-k locations and gather X_k, $\tilde{X}_k$, and S_k. Stage 2 computes cross-attention from all N queries to only the selected k keys and values, where logits are modulated by $\tilde{X}_k$ and values are gated by S_k. Standard and CGA are blended to produce the final aggregation with complexity $O(Nk)$.

3) Curvature Modulation and Blending: We broadcast $\tilde{X}_c$ to the logit layout and modulate the logits as follows:

$$\tilde{L} = L_{\text{Std}} \odot \left(1 + \alpha \odot \tilde{X}_c\right), \quad A_{\text{Curv}} = \mathcal{H}_{\text{Softmax}}(\tilde{L})$$

where $\odot$ denotes the elementwise multiplication. This multiplicative form increases logits around curvature-salient neighborhoods while keeping behavior close to the baseline when $\tilde{X}_c$ is small, which helps stability in flat/nonridge regions. We then compute the convex blend

$$A_{\text{Blend}} = \beta \odot A_{\text{Std}} + (1 - \beta) \odot A_{\text{Curv}}, \quad \beta \in [0, 1]$$

which is applied to the values to obtain X_{out} (reshaped back to the window layout). The learnable β allows LCGA to emphasize ridge neighborhoods when curvature evidence is strong, while preserving the standard attention behavior elsewhere.

C. Curvature-Guided Token Attention

CGTA performs curvature-guided global association over the full-feature map $X_{\text{in}} \in \mathbb{R}^{H \times W \times C}$ with $N = H \cdot W$ tokens, while avoiding the dense global attention. As shown in Fig. 4, CGTA uses a two-stage routing design: it first selects a compact top-k token set guided by curvature-aware saliency and then computes blended cross attention only over the selected keys/values, reducing the dominant cost to $O(Nk)$ with $k \ll N$.

1) Stage 1 (Curvature-Aware Token Selection): We reuse the LCGA proxy densely to produce $\tilde{X}_c \in \mathbb{R}^{H \times W}$, as illustrated in Fig. 4. In parallel, a retention gate produces a scalar reliability map $X_{\text{gs}} \in [0, 1]$, which helps to suppress unstable activations and prevents selection from being dominated by noisy high-response locations. The two cues are combined into a single score

$$\text{Score} = \frac{1}{2}\left(|\tilde{X}_c| + X_{\text{gs}}\right).$$

Selection Score and Budget: A size-aware KTokens policy shown in Fig. 4 determines the routing density ρ from (H, W) and sets the token budget as follows:

$$\begin{aligned} \texttt{time\_level} &= \max\left(\text{int}\left(\log\left(\frac{H}{r}\right)\right), \text{int}\left(\log\left(\frac{W}{r}\right)\right)\right) \\ \rho &= 4^{\texttt{time\_level}} \\ k &= \text{int}\left(\frac{H}{\rho}\right) \times \text{int}\left(\frac{W}{\rho}\right) \end{aligned} \tag{1}$$

where r represents the reduction factor. Here, $k \ll N$ increases sublinearly as the image grows, keeping the dominant cross-attention cost $O(Nk)$ near-linear while preserving informative context. Let $\tilde{S}_k = \mathcal{H}_{\text{Top-}k}(\text{Score})$ be the index set of the k highest scoring locations. We then use $\mathcal{H}_{\text{Gather}}(\cdot)$ (an index-based gather operator that extracts rows/elements at the selected indices) to obtain

$$X_k \in \mathbb{R}^{k \times C}, \quad \tilde{X}_k \in \mathbb{R}^{k \times 1}, \; S_k \in \mathbb{R}^{k \times 1}.$$

Here, X_k provides the selected token features for global association, $\tilde{X}_k$ supplies curvature saliency to modulate attention logits in Stage 2, and S_k carries the selection confidence to gate values; we carry all three together with X_{in} into Stage 2.

TABLE I
DATASET SUMMARY USED IN OUR EXPERIMENTS

Properties	Train	Validation	Test		
Data Name	AID [37]	AID [37]	AID [37]	UCMerced [38]	WHU-RS19 [39]
Used Images	7850 (10000)	150 (10000)	2000 (10000)	1050 (2100)	1002
Original Size	600×600	600×600	600×600	256×256	600 × 600
Spatial Resolution	0.5∼8 m	0.5∼8 m	0.5∼8 m	0.3 m	—
Categories	30	30	30	21	19
Task	Scene Classification	Scene Classification	Scene Classification		

2) Stage 2 (Blended Cross-Attention Over Selected Tokens): We compute queries for all N locations with a linear projection of reduced width $c = (C/2)$

$$Q = \mathcal{H}_{\text{Lin}}(X_{\text{in}}) \in \mathbb{R}^{N\times c}.$$

For the k selected tokens, we first compress channels with a 1×1 convolution and then apply linear heads to obtain

$$K \in \mathbb{R}^{k\times c}, \quad \text{and } V \in \mathbb{R}^{k\times C}$$

so the inner dimension of $Q \otimes K^T$ is c (reduced for efficiency), while V keeps width C to preserve synthesis capacity. We gate values by selection confidence

$$\tilde{V} = V \odot S_k$$

so weakly selected tokens contribute less to aggregation. The standard and curvature-modulated logits over the reduced key set are

$$L_{\text{Std}} = \frac{Q \otimes K^T}{\sqrt{c}}, \quad \tilde{L} = L_{\text{Std}} \odot \left(1 + \alpha \odot \tilde{X}_k\right)$$

with

$$A_{\text{Std}} = \mathcal{H}_{\text{Softmax}}(L_{\text{Std}}), \quad A_{\text{Curv}} = \mathcal{H}_{\text{Softmax}}(\tilde{L})$$

and blended attention

$$A_{\text{Blend}} = \beta \odot A_{\text{Std}} + (1-\beta) \odot A_{\text{Curv}}.$$

Aggregation with the gated values yields tokens $A_{\text{Blend}} \otimes \tilde{V}$. Because attention is computed from N queries to only k keys/values ($k \ll N$), the dominant cost scales as $O(Nk)$, enabling efficient long-range association guided by the curvilinear saliency.

D. Reconstruction

In the final stage, a lightweight reconstruction head $\mathcal{H}_{RC}$ maps the fused features $\mathcal{F}_0 + \mathcal{F}_{\text{DF}}$ back to RGB at the target scale. Specifically, we first project features with a 3×3 convolution and then upsample via a subpixel rearrangement to produce $I_{\text{SR}} \in \mathbb{R}^{H'\times W'\times 3}$; a single 3×3 refinement convolution is applied after upsampling. This head is intentionally low-capacity and largely linear: PixelShuffle is a deterministic rearrangement of feature channels and does not synthesize new content on its own, while the two shallow convolutions mainly perform feature-to-RGB mapping and mild refinement. As a result, fine details in the output are primarily determined by the backbone features shaped by LCGA/CGTA under supervised reconstruction, rather than by an expressive decoder that could invent textures. Moreover, we train CGA with an L_1 fidelity objective, which penalizes deviations from HR references and discourages visually convincing but incorrect patterns. In contrast to GAN-based perceptual SR pipelines that explicitly encourage sharp textures and may introduce details that are not supported by the LR observation, our reconstruction emphasizes fidelity-oriented reconstruction with controlled complexity, thereby reducing the tendency to introduce hallucinated artifacts.

IV. EXPERIMENTS

A. RS Datasets

We evaluate on three public RS datasets: AID [37], UCMerced [38], and WHU-RS19 [39]. A summary of the splits used in our study (training/validation/testing) is provided in Table I. For supervision, we construct paired HR–LR data per dataset by treating the native images as HR references and generating LR inputs via bicubic downsampling at the target scale. Unless otherwise noted, all settings are identical across datasets.

B. Implementation Details

1) Model Details: We instantiate two variants for $\times 2$ and $\times 4$ SR: CGA and a larger capacity CGA-L. Following [34], both use six residual groups (RGs), each containing six curvature-guided attention blocks (CGABs). The number of attention heads and the MLP expansion ratio are set to 6 and 2, respectively. The window size is 8×32. The channel width is $C = 180$ for CGA and $C = 240$ for CGA-L.

2) Training Details: Models are trained with mini-batches of size 4. Each LR input is a random 64×64 crop; we train for 5×10^5 iterations with random horizontal flips and rotations by 90°, 180°, and 270°. We optimize the $\mathcal{L}_1$ loss using Adam with $\beta_1 = 0.9$ and $\beta_2 = 0.99$. The initial learning rate is 2×10^{-4} and is halved at 250k, 400k, 450k, and 475k iterations. All models are implemented in PyTorch and trained on a single NVIDIA RTX 3090 GPU.

3) Evaluation Metrics: We evaluate SR quality using six evaluation metrics covering pixel, structural, spectral, and perceptual aspects: PSNR, SSIM, VIF, and SCC (higher is better), and SAM and FID (lower is better) [7], [40]. Together, these metrics provide a broader assessment than pixelwise measures alone.

C. Ablation Studies

We conduct controlled ablations on UCMerced 64×64 LR images at $\times 4$ SR (945 training/1050 testing images). Unless

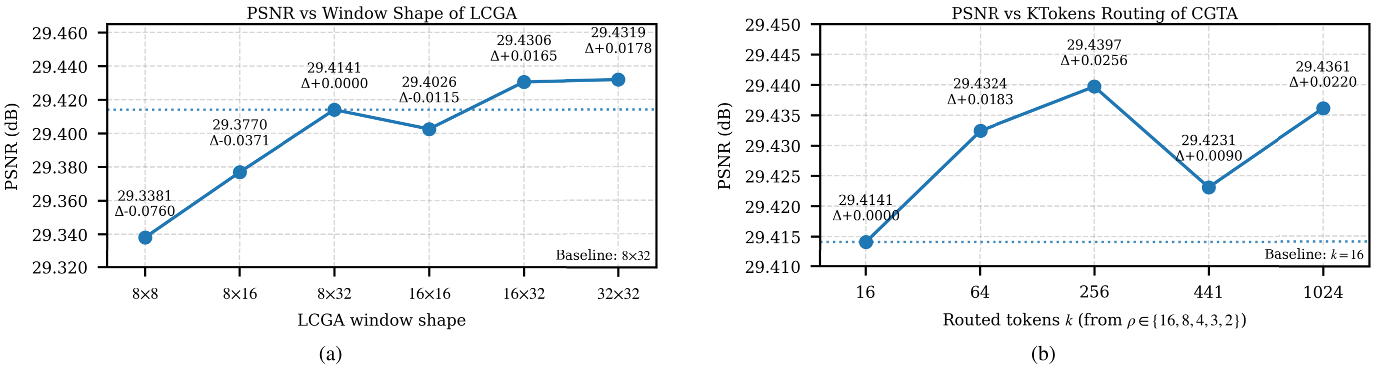


Fig. 5. Ablation results on UCMerced at ×4 upscaling. (a) LCGA window shape while keeping the CGTA routing fixed to the baseline KTokens setting $\rho = 16$ and $k = 16$. (b) CGTA routing density by varying $\rho \in \{16, 8, 4, 3, 2\}$ which changes the routed token budget k while keeping the LCGA window fixed to 8×32. The baseline settings are indicated in the plots. Each point reports PSNR in decibels and the relative change Δ with respect to the baseline.

noted, all settings follow Section IV-B2; for this smaller corpus, we train for 80k iterations with milestones at {30k, 50k, 65k, 70k, 75k} following [41]. Each ablation varies a single factor while keeping others fixed. We first analyze module and design choices (LCGA/CGTA/Blend toggles, projection dimension d, window, and token budgets) under a common configuration and then examine why blending helps using the ridge-band diagnostic introduced in the Section I.

1) LCGA Window Shape: Fig. 5(a) studies the effect of the LCGA window shape using 64×64 size images for both training and testing. Under this fixed setting, the CGTA routing configuration remains unchanged: with the baseline KTokens schedule presented in (1), 64×64 yields $\rho = 16$ and thus $k = 16$. PSNR increases as the window becomes larger, but the gain saturates quickly: the default 8×32 window achieves 29.4141 dB, while the best 32×32 window reaches 29.4319 dB, i.e., only +0.0178 dB. Since enlarging the window expands the local attention region and increases FLOPs and activation memory, we use 8×32 as the default and suggest enlarging it only when additional compute is available and accuracy is the primary objective.

2) CGTA Routing Density: Fig. 5(b) evaluates how CGTA behaves under different global routing densities, again on 64×64 patches for both training and testing. In CGTA, the routed token count is determined by the KTokens rule as $k = \lfloor H/\rho \rfloor \lfloor W/\rho \rfloor$, where ρ controls routing density. For this ablation, we fix $\rho \in \{16, 8, 4, 3, 2\}$, which on 64×64 corresponds to $k \in \{16, 64, 256, 441, 1024\}$. PSNR changes slightly as routing becomes denser and then saturates: the baseline ($\rho = 16$, $k = 16$) gives 29.4141 dB, while $\rho = 8$ ($k = 64$) and $\rho = 4$ ($k = 256$) reach 29.4324 dB (+ 0.0183) and 29.4397 dB (+ 0.0256); larger budgets do not yield consistent improvement. Because the dominant cross-token interaction scales as $O(Nk)$, smaller ρ (larger k) increases computation approximately linearly with routed tokens, so lightweight routing provides a favorable accuracy–efficiency tradeoff.

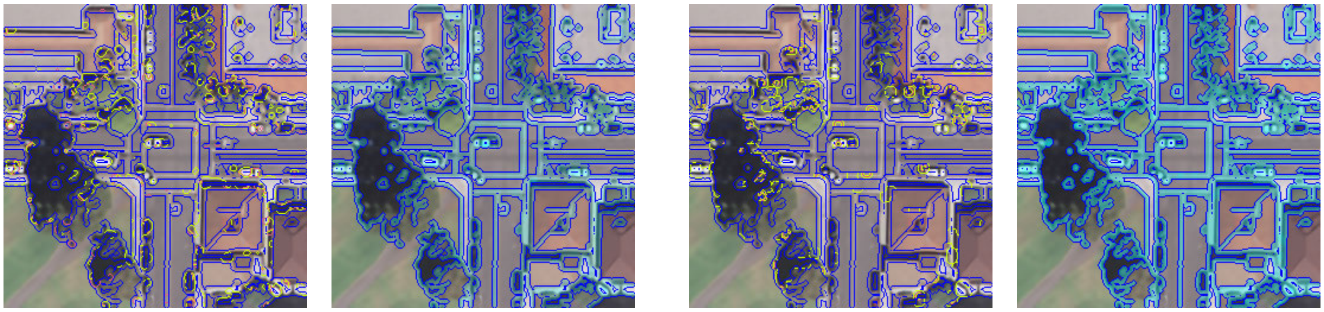
3) Module and Design Ablations (8×32, $k = 16$): Table II summarizes the component effects under the common configuration (8×32 window and $k = 16$ on 64×64). Removing LCGA produces the largest drop (29.1993, −0.2148 dB), evidencing its impact on ridge-aware local aggregation; removing CGTA yields a smaller but consistent reduction (29.3900, −0.0241 dB), reflecting its complementary global role. Disabling blending reduces PSNR (29.3837, −0.0304 dB), confirming that curvature-standard mixing stabilizes attention away from ridge neighborhoods. Increasing the CGTA query projection from $d = C/2$ to $d = C$ gives only a minor gain (29.4165, +0.0024 dB), so we keep $d = C/2$ to reduce the cost of $QK^\top$ while preserving C-dimensional expressiveness in V.

TABLE II
COMPONENT ABLATION RESULTS ON UCMERCED AT ×4

LCGA	CGTA	Blend	d	PSNR (dB)	Δ vs Full	Params (M)	FLOPs (G)
✗	✓	✓	$C/2$	29.1993	−0.2148	10.03	36.79
✓	✗	✓	–	29.3900	−0.0241	10.90	59.50
✓	✓	✗	$C/2$	29.3837	−0.0304	10.11	46.95
✓	✓	✓	C	29.4165	+0.0024	11.79	49.47
✓	✓	✓	$C/2$	29.4141	–	10.47	48.15

Choice for Main Experiments and Practical Hyperparameter Selection: Across Fig. 5(a) and (b) and Table II, more expensive settings typically yield <0.03-dB additional PSNR on fixed-size patches, while cost rises substantially with larger windows or denser routing. We, therefore, adopt a single default configuration in the main model: LCGA window 8×32, $d = C/2$, and the baseline KTokens routing. In practical use, the test image size may vary with the dataset and scale factor; accordingly, k is not kept fixed. Instead, the KTokens schedule computes ρ from (H, W) and adapts k automatically: it reduces to $k = 16$ on 64×64 patches, and selects more tokens on larger tiles while keeping $k \ll N$ so the dominant attention cost remains near-linear ($O(Nk)$). Practitioners can start from the default; under tight memory or latency constraints, keep the same window and routing schedule and reduce channel-related cost, whereas for accuracy-critical scenarios with sufficient compute, modestly enlarging the window or using denser routing can provide small gains with diminishing returns.

4) Why Blending Matters (Ridge-Band Attention Ablation): We reuse the ridge-band diagnostic introduced in the Introduction (RBAR/RAG) to assess how attention concentrates on edge- and layout-bearing neighborhoods. We evaluate three



(a) (b) (c) (d)

(a) and (b) represents LCGA (c) and (d) represents CGTA

Fig. 6. Ridge-band attention on intersection95 from UCMerced at ×4 scale. Blue outlines indicate the HR ridge band. (Left) Top-dominance contours that compare Std and Curv attention. (Right) Blend attention heatmap within the same band. Curv frequently dominates along road branches and around junction boundaries, while blending concentrates attention on these ridge neighborhoods and remains stable. Please see zoomed-in view for better view. (a) Top-dominance (Std versus Curv). (b) Blend overlay. (c) Top-dominance (Std versus Curv). (d) Blend overlay. (a) and (b) represents LCGA. (c) and (d) represents CGTA.

TABLE III

Ridge-Band Attention Ablation on UCMerced at ×4 With Values Averaged Over the Validation Set

Module	**RBAR (Std)**	**RBAR (Curv)**	**RBAR (Blend)**	**RAG (Blend−Std)**
LCGA	0.3588	0.3592	0.3590	+0.0002
CGTA	0.3602	0.3602	0.3602	0.0000

variants for both LCGA and CGTA: the standard attention (Std), the curvature-modulated attention (Curv), and their blended form (Blend) and report a RAG as Blend–Std. On UCMerced at ×4, the dataset means are summarized in Table III, and a representative case is visualized in Fig. 6.

The mean results in Table III indicate that Blend allocates slightly more mass to the ridge band than Std for LCGA and closely tracks Curv, yielding a small positive gain. For CGTA, the mean RAG is approximately 0; a contributing factor is the compact global token budget ($k = 16$), which keeps selection conservative so that blending preserves curvature-induced emphasis without pushing additional mass that could over-concentrate or drift. In the visualization of Fig. 6, Curv frequently dominates along road arms and junction boundaries (top-dominance contours), while Blend concentrates attention on these ridge neighborhoods in a stable manner (overlay), supporting the intended role of blending in improving ridge-sensitive fidelity without additional branches or heavyweight global modules.

D. Comparison With the State-of-the-Art Methods

We compare CGA against bicubic interpolation and recent RS image SR baselines: HSENET [42], TransENet [43], HAUNet [15], TTST [17], ACTSR [44], and HMoE [41]. All models and CGA are retrained from the authors' open-source code.

Table IV reports the peak GPU memory, parameter count, FLOPs, per-image latency, and PSNR on UCMerced ×4. CGA attains a superior cost–accuracy tradeoff: with only 10.47M parameters and 48.15G FLOPs, it achieves 29.77-dB result in exceeding TTST (29.69 dB) and HMoE (29.71 dB) while using less memory and/or fewer FLOPs. The larger CGA-L variant yields the highest PSNR (29.81 dB) with moderate overhead. These results align with our design goal: curvature-guided local–global attention improves ridge-sensitive fidelity without incurring high computational cost.

TABLE IV

Comparison of Different Models on UCMerced ×4 SR Test Dataset. Best and Second Best PSNR Are Highlighted in Red and Blue, Respectively

Method	**Mem.** (MB)	**Params** (M)	**FLOPs** (G)	**Latency** (ms)	**PSNR** (dB)
Bicubic	—	—	—	—	26.9128
HSENET [42]	1668	5.43	19.20	29.20	29.4844
TransENet [43]	734	37.46	21.44	27.63	29.6181
HAUNet [15]	2522	2.63	6.18	48.25	29.7324
TTST [17]	84	18.37	85.32	185.64	29.6910
ACTSR [44]	1844	4.44	20.59	88.43	29.6859
HMoE [41]	120	27.09	128.11	141.63	29.7524
CGA (ours)	152	10.47	48.15	126.41	29.7757
CGA-L (ours)	176	18.20	79.62	138.21	29.8078

1) Quantitative Results: Table V reports six complementary metrics (PSNR/SSIM/VIF/SCC/SAM/FID) on AID, UCMerced, and WHU-RS19 at ×2 and ×4. Across datasets and scales, CGA and CGA-L consistently achieve top or near-top results, indicating improvements that are not limited to a single evaluation criterion. In addition to higher pixelwise fidelity (PSNR/SSIM), the gains are reflected in higher VIF and SCC, suggesting improved structural consistency in the reconstructed content. Meanwhile, the lower SAM values indicate reduced cross-channel distortion, which is particularly relevant for remote-sensing imagery where preserving spectral relationships is important. Finally, the reduced FID values indicate a smaller distributional gap to HR references in perceptual feature space, which is consistent with the improved perceptual realism.

To complement the dataset-level averages in Table V, Fig. 7(a) and (b) reports per-class PSNR ranking behavior at ×4 on AID and UCMerced, respectively. The CGA

TABLE V

QUANTITATIVE COMPARISON ON UCMERCED, AID, AND WHU DATASETS. BEST AND SECOND BEST RESULTS ARE COLORED IN RED AND BLUE, RESPECTIVELY

Dataset	Scale	Metric	Bicubic	HSENET [42]	TransENet [43]	HAUNet [15]	TTST [17]	ACTSR [44]	HMoE [41]	CGA (ours)	CGA-L (ours)
AID	2	PSNR↑	33.8429	37.0641	36.9148	37.0896	37.1784	37.2617	37.2187	37.2877	37.2967
		SSIM↑	0.9019	0.9458	0.9443	0.9459	0.9468	0.9476	0.9474	0.9477	0.9479
		VIF↑	0.6069	0.7687	0.7626	0.7869	0.7726	0.7756	0.7763	0.7761	0.7767
		SCC↑	0.5202	0.6627	0.6538	0.6616	0.6670	0.6716	0.6672	0.6748	0.6760
		SAM↓	0.0742	0.0520	0.0530	0.0518	0.0513	0.0508	0.0510	0.0507	0.0506
		FID↓	5.5224	0.3593	0.3859	0.3567	0.3455	0.3278	0.3283	0.3249	0.3200
	4	PSNR↑	28.6838	30.8095	30.8758	30.9260	30.9543	30.9940	31.0030	31.0504	31.0689
		SSIM↑	0.7333	0.8119	0.8142	0.8153	0.8165	0.8177	0.8179	0.8192	0.8195
		VIF↑	0.2510	0.3658	0.3674	0.3703	0.3726	0.3751	0.3760	0.3774	0.3782
		SCC↑	0.1416	0.2829	0.2883	0.2936	0.2943	0.2952	0.2958	0.2991	0.2997
		SAM↓	0.1275	0.1010	0.1002	0.0997	0.0993	0.0989	0.0988	0.0983	0.0981
		FID↓	34.3511	20.9356	20.4833	20.2602	20.3157	20.3698	20.3071	20.1585	20.0311
UCMerced	2	PSNR↑	32.0895	36.3028	36.0315	36.2644	36.4259	36.4686	36.5097	36.5204	36.5499
		SSIM↑	0.8925	0.9459	0.9438	0.9457	0.9472	0.9478	0.9479	0.9480	0.9483
		VIF↑	0.5774	0.7707	0.7606	0.7695	0.7745	0.7762	7769	0.7773	0.7797
		SCC↑	0.4968	0.6494	0.6405	0.6488	0.6550	0.6566	0.6589	0.6583	0.6588
		SAM↓	0.0736	0.0480	0.0492	0.0478	0.0472	0.0470	0.0468	0.0467	0.0466
		FID↓	41.2732	26.4416	27.3029	25.5044	25.9316	25.5708	25.5371	25.5044	25.3684
	4	PSNR↑	26.9128	29.4844	29.6181	29.7324	29.6910	29.6859	29.7524	29.7757	29.8078
		SSIM↑	0.7008	0.7977	0.8022	0.8054	0.8050	0.8057	0.8055	0.8073	0.8082
		VIF↑	0.2301	0.3636	0.3666	0.3729	0.3730	0.3734	0.3767	0.3767	0.3781
		SCC↑	0.1425	0.2964	0.3048	0.3120	0.3114	0.3113	0.3132	0.3166	0.3183
		SAM↓	0.1287	0.0996	0.0977	0.0965	0.0970	0.0971	0.0966	0.0963	0.0960
		FID↓	111.2555	98.9639	99.2305	96.1736	99.2459	98.6537	98.3644	97.4297	94.5828
WHU-RS19	2	PSNR↑	31.8255	31.9049	31.9739	31.9708	31.9356	31.9617	31.9476	31.9754	31.9779
		SSIM↑	0.8445	0.8465	0.8480	0.8477	0.8471	0.8479	0.8477	0.8483	0.8480
		VIF↑	0.3942	0.4133	0.4137	0.4144	0.4140	0.4148	0.4149	0.4150	0.4148
		SCC↑	0.2795	0.3048	0.3086	0.3080	0.3075	0.3089	0.3056	0.3093	0.3094
		SAM↓	0.0877	0.0863	0.0857	0.0857	0.0860	0.0858	0.0859	0.0857	0.0860
		FID↓	22.0019	21.9990	21.9340	21.8856	21.5922	21.5721	21.5142	21.3693	19.5362
	4	PSNR↑	27.8284	28.0690	28.1087	28.1154	28.0645	28.1019	28.1072	28.1181	28.1203
		SSIM↑	0.6812	0.6973	0.6987	0.6989	0.6970	0.6988	0.6986	0.6989	0.6990
		VIF↑	0.1517	0.1665	0.1674	0.1678	0.1667	0.1676	0.1676	0.1679	0.1678
		SCC↑	0.0766	0.1194	0.1222	0.1224	0.1200	0.1226	0.1226	0.1236	0.1236
		SAM↓	0.1392	0.1347	0.1341	0.1339	0.1348	0.1342	0.1341	0.1342	0.1341
		FID↓	57.0156	56.3537	56.9616	56.6437	56.9871	56.7239	56.4465	56.2450	56.1721

family maintains strong ranks across a broad range of scene categories rather than relying on improvements concentrated in a small subset of classes, indicating stable generalization across diverse aerial content. This classwise view is consistent with the overall trends in Table V and supports that the proposed curvature-guided local–global modeling benefits varied structures (including thin boundaries and curvilinear patterns). Additional classwise results are provided in the supplementary material.

Overall, the improvements are consistent across pixelwise (PSNR/SSIM), structural (VIF/SCC), spectral (SAM), and perceptual/distributional (FID) measures, corroborating the effectiveness of the curvature-guided local–global design. The gains are driven in part by ridge-sensitive fidelity, including improved continuity and alignment of linear structures and more stable reconstruction of periodic textures, which elevates full-reference scores without requiring aggressive texture amplification. At the same time, the adopted window and routing settings follow the accuracy–efficiency tradeoffs discussed in the ablations, enabling these gains with controlled computational cost.

2) Qualitative Results: Fig. 8 presents the visual comparisons at $\times 4$ on three representative cases: AID playground_141 (court-divider stripe), UCMerced buildings95 (roof patch), and WHU-RS19 airport_12 (parking/shed area). Across these scenes, CGA-L and CGA produce reconstructions that are visually closest to the HR reference, with clearer structure, more similar textures, and noticeably fewer artifacts, while other approaches tend to yield softer geometry and less consistent texture under the same degradation.

It can be observed that CGA-L recovers the divider stripe in playground_141 with continuous, well-aligned geometry; on buildings95, CGA and CGA-L reconstruct the roof region with more regular texture, improved perceptual realism, and surface continuity; and in airport_12, CGA-L preserves the layout and boundaries of the parking and shed areas, with CGA offering comparable visual quality. These observations align with the intended effect of the proposed attention enhanced ridge-sensitive fidelity where, reinforcing curvilinear continuity and texture regularity suppresses spurious patterns and reduces aliasing, leading to better SR outputs at challenging $\times 4$ magnification.

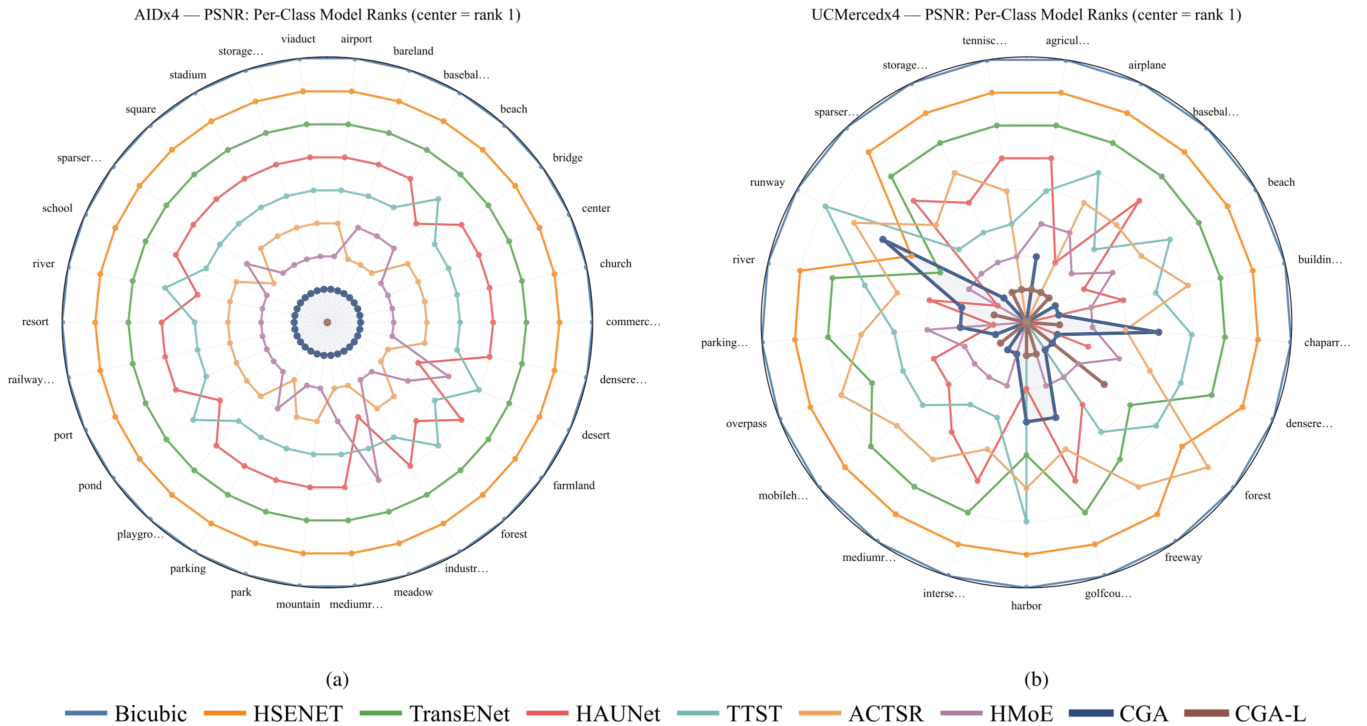


Fig. 7. Per-class rank radar analysis at $\times 4$ upscaling. (a) and (b) PSNR ranks on AID and UCMerced. The center indicates rank 1 so a smaller radius is better. Each method is shown as a radar trace and the traces of CGA and CGA-L are drawn with slightly thicker lines than the others. Please see zoomed-in view for a better view.

TABLE VI
SMALL OBJECT DETECTION ON DOTAv1. BEST RESULTS ARE IN RED

Method	PSNR (dB)	SSIM	Plane AP	Helicopter AP	Ship AP	Small Vehicle AP	mAP
TTST	34.1368	0.8428	0.635	0.714	0.840	0.512	0.675
ACTSR	34.1620	0.8434	0.637	0.738	0.843	0.595	0.703
HMoE	34.1488	0.8431	0.637	0.704	0.842	0.596	0.695
CGA (ours)	34.1794	0.8438	0.629	0.829	0.867	0.703	0.757

E. Additional Task: Small Object Detection

To evaluate whether SR improvements transfer to downstream recognition, we conduct a small-object oriented detection study on DOTAv1 [45] using the YOLO11n-obb detector [46]. We use the official YOLO11n-obb model pretrained on the DOTAv1 training set (1411 images and 15 classes) and keep the detector fixed for all comparisons (no fine-tuning on SR outputs), so that performance differences primarily reflect SR quality rather than detector retraining.

We follow a controlled SR-to-detection pipeline. First, LR inputs are generated from validation images via bicubic $\times 4$ downscaling. To ensure an exact spatial correspondence between the recovered SR outputs and the original HR images (so that both full-reference SR metrics and detection are evaluated on aligned images), we evaluate on the subset of the DOTAv1 validation split whose height and width are divisible by 4, which yields 89 validation images. Each SR method [TTST [17], ACTSR [44], HMoE [41], and CGA (ours)] super-resolves the same LR set, and the resulting SR images are fed to the same pretrained detector under identical evaluation settings. We report average precision (AP) per class and mean average precision (mAP) at intersection-over-union (IoU) = 0.5.

As summarized in Table VI, CGA attains the highest mAP and leads most categories, indicating improved recovery of structure and local detail that benefits oriented detection. Qualitatively presented in Fig. 9, on ship crops several baselines miss targets that are correctly detected after CGA, and on dense plane layouts some baselines merge two nearby objects into a single box whereas CGA separates them. These patterns are consistent with our design goal: curvature-guided local–global attention improves ridge-sensitive fidelity, yielding sharper, more continuous contours and more regular textures on small objects, which in turn reduces missed detections and mitigates box merging in crowded or elongated targets. For completeness, full results over all 15 categories and the corresponding precision–recall curves are provided in the supplementary material.

F. Additional Task: Experiment on Real-World Data

To verify the effectiveness of the proposed CGA under real-world degradations, we conduct experiments on the

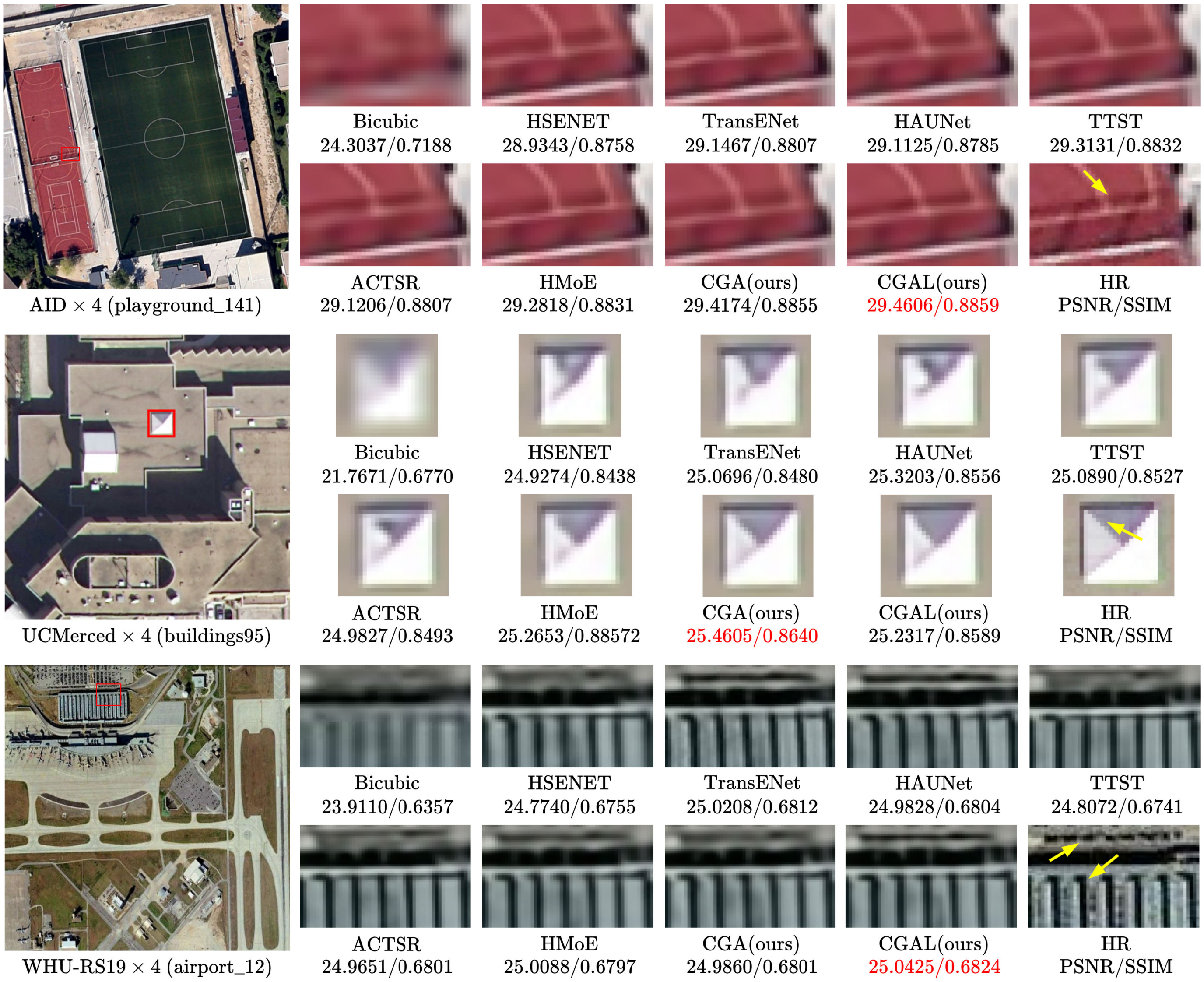


Fig. 8. SR ×4 visual comparisons on different datasets with different methods. The best PSNR/SSIM is shown in red. Please see zoomed-in view for a better view.

NWPU-RESISC45 dataset [47] without applying any simulated degradation. We randomly sample a class-balanced subset of 25 images per class (45 classes), resulting in 1125 images in total, and evaluate all methods under the same protocol.

1) Quantitative Results: Following common practice for real-world evaluation without ground-truth HR references, we report NIQE (lower is better) together with three learned perceptual quality metrics, i.e., CLIPIQA, MUSIQ, and MANIQA (higher is better) [40]. As shown in Table VII, CGA achieves the best NIQE (7.0486) and also attains the highest MUSIQ (35.7594) and MANIQA (0.2421), while remaining competitive on CLIPIQA. These results indicate that CGA improves perceptual quality on real-world imagery and generalizes beyond the simulated degradation setting used in standard SR benchmarks, consistent with a reduced tendency to introduce unnatural artifacts.

2) Qualitative Results: Fig. 10 provides qualitative comparisons on a representative real-world RS sample with two zoomed-in-view regions. In the first zoom (blue box, top row), most competing methods introduce noticeable noise and fail to recover the thin markings, producing broken or smeared line structures, whereas CGA reconstructs clearer and more continuous lines with fewer spurious pixels. In the second zoom (red box, bottom row), the curved outline is often

TABLE VII
REAL-WORLD COMPARISON ON NWPU-RESISC45 (×4). BEST AND SECOND-BEST RESULTS ARE HIGHLIGHTED IN RED AND BLUE, RESPECTIVELY

Method	NIQE↓	CLIPIQA↑	MUSIQ↑	MANIQA↑
HSENET [42]	7.1749	0.1796	35.4215	0.2302
TransENet [43]	7.4072	0.1794	34.6675	0.2297
HAUNet [15]	7.3460	0.1820	34.8294	0.2311
TTST [17]	7.2768	0.1862	35.1091	0.2301
ACTSR [44]	7.5590	0.4671	35.6359	0.2408
HMoE [41]	7.5966	0.4461	35.3918	0.2370
CGA (ours)	7.0486	0.4636	35.7594	0.2421

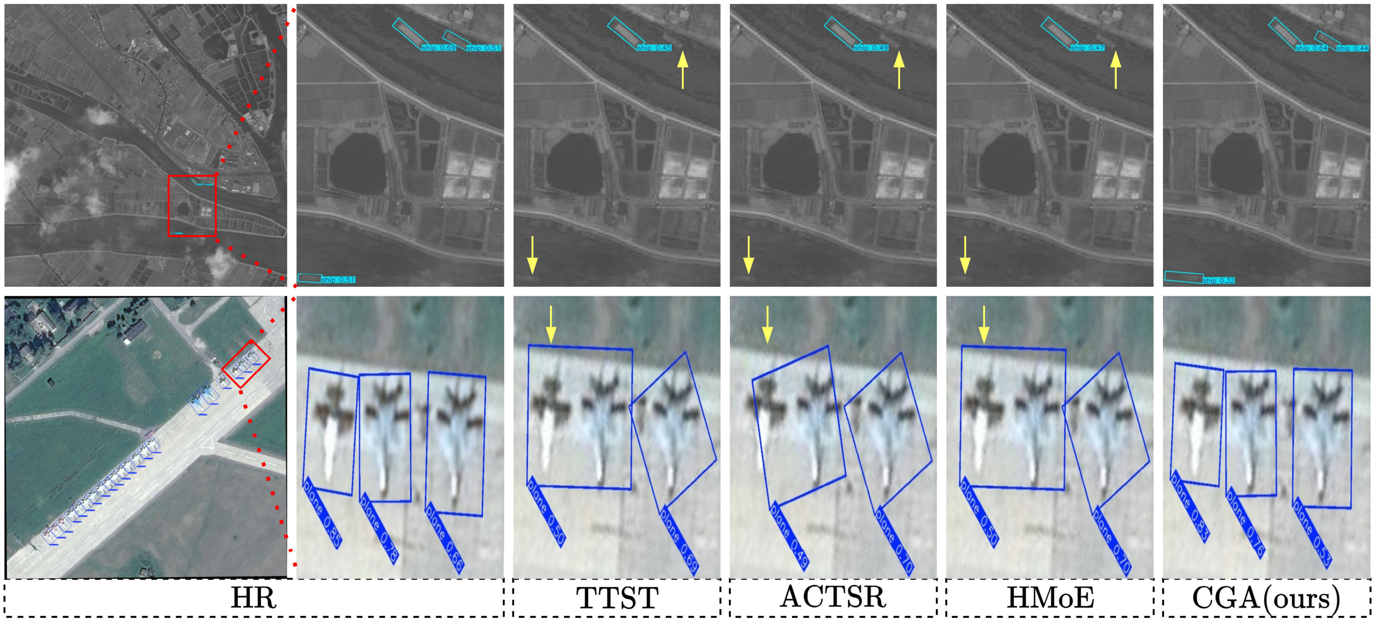


Fig. 9. SR ×4 visualization results on the DOTAv1 dataset for the downstream object detection setting. This figure compares different SR methods and highlights how reconstruction quality affects small objects and boundary details in the zoomed-in-view regions. Please see zoomed-in view for a better view.

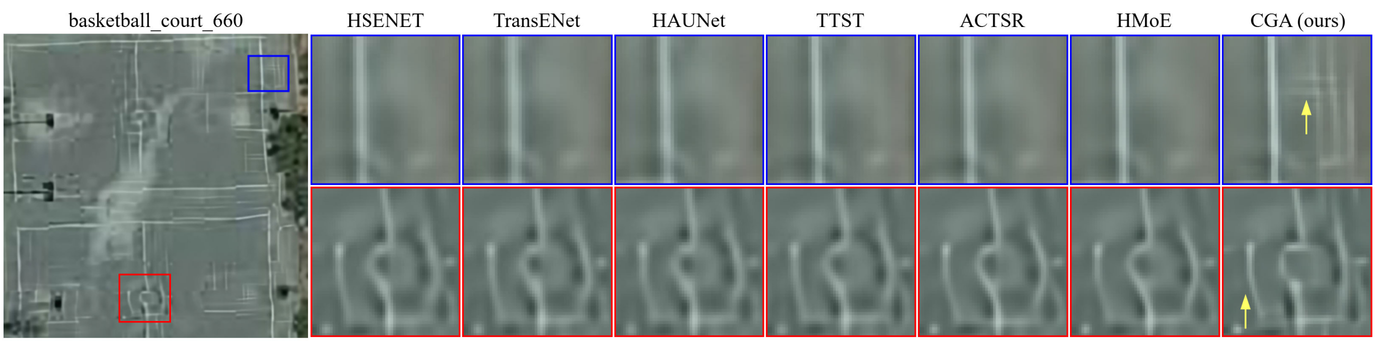


Fig. 10. SR ×4 visualization comparison of various methods on NWPU-RESISC45 with real-world degradation.

rendered with blurred, spread-out strokes by other methods, while CGA yields a sharper and cleaner curve with better pixel compactness and continuity. Overall, CGA produces more visually pleasing textures and more realistic structural outlines, reducing noise and blurring artifacts and demonstrating practical applicability under real-world degradations.

V. Discussion

The CGA family aims to improve the ridge-sensitive fidelity in SR by combining curvature-guided local aggregation with selective global routing. The main quantitative comparisons show that CGA and CGA-L achieve consistently strong performance across six complementary metrics, indicating that the gains are not limited to pixelwise measures but also reflect improved structural and spectral consistency, and closer alignment to the HR reference distribution in perceptual feature space. From a practical standpoint, the efficiency comparison supports that these gains can be achieved with controlled computational overhead and a favorable accuracy–efficiency tradeoff.

Beyond dataset-level averages, the classwise analysis at ×4 indicates stable behavior across scene categories. The CGA family maintains strong per-class PSNR ranks on AID and UCMerced, suggesting that improvements are broadly distributed rather than concentrated in a small subset of classes. Qualitative comparisons are consistent with this trend, showing clearer boundary continuity and more stable textures under challenging magnification. In addition, the detection results indicate that using CGA/CGA-L outputs can improve small-object detection performance, suggesting that the recovered details are informative for downstream analytics. Consistently, the real-world qualitative comparisons also show cleaner structural outlines with fewer noise/blurring artifacts, aligning with the improvements reported by no-reference perceptual metrics.

The ridge-band diagnostics and blending ablations (see Table III and Fig. 6) further support the design rationale that blending stabilizes ridge-focused attention and complements the curvature-guided pathway without relying on heavy global computation.

Limitations remain. First, gains may be smaller for scenes dominated by large homogeneous regions or weak curvilinear structure, where ridge guidance is less informative. Second, although the default window and routing settings provide a favorable tradeoff, increasing window size or routing density can raise cost for only marginal improvements, so the accuracy–efficiency balance should be selected according to deployment constraints. Third, as with most learning-based SR methods, the model may be sensitive to dataset shift, and additional adaptation may be beneficial for unseen sensors or acquisition conditions.

Overall, the evidence across multimetric benchmarks, classwise stability, qualitative examples, downstream detection, and real-world evaluation supports the effectiveness of the curvature-guided local–global design while maintaining a practical compute profile.

VI. Conclusion

Empirically, CGA achieves cross-dataset state-of-the-art overall averages in computation metrics, exhibits consistent per-class superiority in rank–radar analyses, and yields tangible downstream gains in small-object detection mAP, while also showing improved perceptual quality in real-world evaluation without simulated degradation. Qualitative comparisons further exhibit clearer geometry with fewer visible artifacts, corroborating the quantitative trends. The lightweight variant preserves these benefits under tighter memory and latency, underscoring the practicality of the design. Overall, the results indicate that curvature guidance provides a principled and effective bias for RS image SR across diverse scenes and content types. Looking ahead, we will explore adaptive curvature modulation, efficiency-oriented designs, and task-aware objectives, together with cross-sensor adaptation, to further strengthen the performance and deployability.

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